



Experiment-centric "Frequentist" Approach

Formulate
Hypothesis
(H_0 , H_a)

Setup
Treatments
and Controls

Collect
Entire
Data

Evaluate
Dataset
Statistics

Model
Selection

Fixed
Multivariate
Inference

Establish
Experimental
Design and
Parameters to
Observe

Hypothesis
Testing and
Confidence
Interval

Data-driven "Probabilistic" Approach

Specify Prior
Distribution
(Initial Beliefs)

Setup
Treatments
and Controls

Posterior
Becomes
New Prior

Evaluate
Posterior
Distribution

Collect
Initial
Data
(Evidence)

Update
Prior
Distribution
(Posterior)

Model
Predictive
Posterior
Distribution

Decision
Based on
Posterior
Mode

Design

Time-dependent
Data Collection

Evaluation

Model
Deployment &
Inference